

CO2-AT1-Assignment1

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QUESTIONS

Q1 ✓
Smart Document Editor using
String, StringBuilder and
StringBuffer
10 marks

Q2 ✓
Hospital Token Management
System using List Interface
10 marks

Q3 ✓
Airport Boarding Queue
Management using Queue
Interface
10 marks

Q4 ✓
Digital Certificate Verification
using Set Interface
10 marks

Q5 ✓
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Q1 Smart Document Editor using String, StringBuilder and StringBuffer

10 marks

Question 1 :

1. Read a paragraph from the user.
2. Count the number of characters, words, and sentences.
3. Replace every occurrence of a given word with another word.
4. Reverse each sentence individually.
5. Maintain editing history using StringBuffer.
6. Construct the final formatted document using StringBuilder.
7. Display the original document, editing history, and final document.

Requirement: Use String methods wherever appropriate. Use StringBuilder for document construction and StringBuffer for edit history.

Your Code

```
1 import java.util.*;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6         String paragraph = sc.nextLine();  
7         int characters = paragraph.length();  
8         String[] words = paragraph.trim().split("\\s+");  
9         int wordCount = words.length;  
10        String[] sentenceArray = paragraph.split("[.!?]+");  
11        int sentenceCount = 0;  
12        for (String s : sentenceArray) {  
13            if (!s.trim().isEmpty())  
14                sentenceCount++;  
15        }  
16    }  
17 }
```

Input

Java is a powerful programming
language. Java is easy to learn.
Java
Python

Output

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5/5 answered

Q2 Hospital Token Management System using List Interface

10 marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
1 import java.util.*;  
2 class Patient {  
3     int id;  
4     String name;  
5     String department;  
6  
7     Patient(int id, String name, String department) {  
8         this.id = id;  
9         this.name = name;  
10        this.department = department;  
11    }  
12 }
```

Input

3
101
Rahul
Cardiology

Output

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5/5 answered

Q3 Airport Boarding Queue Management using Queue Interface

10 marks

Airport Boarding Queue Management using Queue Interface

Add passengers to the boarding queue.

Display the current queue.

Board passengers in FIFO order.

Display the next passenger without removing them.

Handle VIP passengers using a priority-based queue.

Display remaining passengers after each boarding operation.

Requirement: Use appropriate Queue implementations.

Your Code

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         Queue<String> queue = new LinkedList<>();
8         PriorityQueue<String> vipQueue = new PriorityQueue<>();
9
10        // Add normal passengers
11        int n = sc.nextInt();
12        sc.nextLine();
```

Input

4
Rahul
Priya
Arun

Output

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10 marks

5/5 answered

Q4 Digital Certificate Verification using Set Interface

10 marks

Digital Certificate Verification using Set Interface

Store certificate IDs using an appropriate Set implementation.

Prevent duplicate certificate entries.

Verify whether a certificate ID exists.

Display certificates in insertion order.

Display certificates in sorted order.

Find the total number of unique certificates issued.

Remove revoked certificates.

Requirement: Use suitable Set implementations.

Your Code

```
1 import java.util.*;  
2  
3 public class Main {  
4     public static void main(String[] args) {  
5         Scanner sc = new Scanner(System.in);  
6  
7         HashSet<String> certificates = new HashSet<>();  
8         int n = sc.nextInt();  
9         sc.nextLine();  
10  
11         for (int i = 0; i < n; i++) {
```

Input

5
C101
C102
C103

Output

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**Q5**

Smart Inventory Management
using Map Interface
10 marks



5/5 answered

Q5 Smart Inventory Management using Map Interface

10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise.

Remove discontinued products.

Display all available products.

Requirement: Use a suitable Map implementation.

Your Code

```
1 import java.util.*;  
2  
3 class Product {  
4     int id;  
5     String name;  
6     String category;  
7     int quantity;  
8     double price;  
9 }
```

Input

```
3  
101  
Laptop  
Electronics  
..
```

Output